MCND training on Image Processing

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Some red tape

- How much do you have hate Programming on a scale of 1-10?
- Have you done some Image-Processing before?

Some of my biases

- I like programming and pretend to be a programmer (which I am not).
- Therefore, I use MATLAB as a programming language, along with terminal tools.
- However, MATLAB can also be used as a GUI app (Ask Arlo) which reduces the effort programming.

Outline for the session

1. Get started !!
2. Image-Processing workflow
3. Application of Concepts
4. Extra Slides

Get started

- Can you guys access the github repo ?
- Add the folder `custom_functions` to PATH.
- I will show some basic principles using a cannonical fluid dynamics problem of **Sedimentation of a solid sphere**.
- Data to be used in the code files is to be found here.

Sedimentation problem

- What we want to know from the experiment?
- Looking into the experiment, what we
  - Can answer from the video?
  - Cannot answer from the video?

Image Processing Workflow

Step-1: Reading Images and Videos

- Try to run the code file - mind you the files are MATLAB Live Script files.

- Any problems?

Step-2: Image Enhancement

- The idea is to **improve** (?) the images.
- I have divided it into three steps -
  - Converting to grayscale
  - Cropping an Image
  - Improving the contrast (Most Important Part !!)
- Any problems ?

Step-3: Image Segmentation

- The idea is to **Extract Features** from the images.
- I have divided it into three steps -
  - Spatial filtering
  - Thresholding
  - Edge detection (Most Important Part !!)
- Any problems ?

Mini project 1

- Can you track the position of the sphere across different frames?
- How does the position vs time look like?
- Does this make sense?

More projects !!

- There is a similar sedimentation experiment with a soft sphere. Can you track the sphere? Will the sphere deform?
- If you are more interested, there is a curling ribbon image from my experiment. Try to detect the edge of the ribbon.

Final Remarks

- This was a **simple** introduction to image processing.
- Image processing is "easy" if your experiments are good (?)
- Once a legend said - *You can be good either in experiments or in image processing*.
- From my experience, the former is better.

Extra Slides

Learning Resources

Following are the useful links to learn MATLAB, Image-processing, Photography, etc.

- Get started with MATLAB - MATLAB Onramp course
- Get started with Image Processing using MATLAB - Image Processing Onramp
- Camera basics - (1) Aperture, (2) Focal length, (3) ISO, (4) Shutter speed
- *Digital Image Processing using MATLAB*, Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddings → book website